

CDS 303 Final Report

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Introductions & Problem Definitions

Many streaming platforms like Spotify have become dependent on data to understand what drives listener engagement. With streaming platforms collecting information on millions and millions of songs, there is an opportunity to identify audio features tied to song popularity. This project focuses on one central question: Which audio features are most closely tied to song popularity? Understanding this relationship between audio features and song popularity matters because popularity is influenced by many factors like marketing, timing, artists' reputation, and more, but musical qualities also play a major role. By analyzing audio attributes, we aim to find certain features correlated with higher popularity scores.

Business Understanding:

Even though the music industry generates massive amounts of information, decisions about songwriting, production, and promotion aren't based on data. To answer the business question, the music industry needs to rely on data insights to predict which audio features are tied to song popularity. We can identify which audio features drive popularity, like danceability, energy, or tempo, and this guides production, promotion, and marketing.

This analysis is useful to several groups within the music industry. It is relevant to producers, artists, researchers, labels, and streaming platforms. Producers and artists can apply the findings during the creation of new songs, using insights to shape the sound and structure of their tracks. Researchers benefit by gaining a clearer understanding of musical and cultural trends. Record labels and streaming platforms can use the results to strengthen playlisting and promotion strategies. The insights are relevant both during song creation and after release, when labels and platforms evaluate how well a track performs. The music industry lacks data-driven insights, and by analyzing songs, we can find features that drive popularity. To answer the business question, we will analyze datasets of Spotify tracks that include a variety of audio features and each song's popularity score. Our approach involves measuring relationships between audio features and popularity, examining correlations to identify patterns, and using our results to determine which features show the strongest ties to popularity.

Literature Review & Prior Work

Business Research:

After extensive research on the papers we reviewed, the main takeaway was how audio features can be used to predict song popularity. These studies compared different models, looked at correlations and showed that features like loudness, danceability, and valence consistently mattered. We used these findings to guide which inputs were kept, how we handled scaling and to confirm that linear regression is an appropriate approach for predicting popularity as a continuous value. Overall, the existing research sites helped validate our feature selection and supported the structure of our modeling process. Figure 1 shows the sources we found, along with a general description of how we implemented them into our project.

One of the papers we found showed the effect of feature selection on the accuracy of music popularity using machine learning algorithms. Authors [Faheem Khan and Ilhan Tarimer](#), among many others, described the effects of feature selection on accurate music popularity classification. The study was conducted using the Spotify dataset, and focused on audio features such as danceability, energy, loudness, etc. This study used multiple modeling approaches (SVM, Random Forest, Decision Tree) to find the class of popularity (high vs low). While we did not use classification, we implemented prioritizing feature selection in our model.

Another study, produced by [Kejun Li from the University of Toronto](#) predicted song popularity in the digital age, also through Spotify's data. This was done using Linear Regression, and the study was completed using 60% training and 40% testing to train its model. From using data from less than ten years and only 2,000 observations for training and test sets, this source goes towards the route us as a group were going for, so we used their analysis as a guide to see how our results could turn out to be similar or differ compared to them, and see whether or not certain audio features have a significant impact in song popularity over others.

[Nijkamp's](#) investigation of the relationship between song popularity and audio features measured the number of streams a song has on Spotify. This study also used a linear regression modeling approach, with the 12 audio features as inputs. This source had a detailed correlation analysis that helped us verify any linear relationships between the target variable and the inputs.

The last source we have, which is also our most recent, dating back to January 2025, summarizes a study using the Indian music market to decode Spotify's hits ([Wukkadada, B. \(2025\)](#)). This study used Spotify API data and exploratory, statistical, and predictive analyses—including ANOVA and sentiment scoring to reveal that popular songs are generally shorter, louder and high in energy. The inputs are similar to the other sources; however, it determines whether those features determine if a song became a hit or not.

Citation	Input/Outputs	Modeling Approach	How we implemented into our model
Khan, F., Tarimer, I., Alwageed, H. S., Karadağ, B. C., Fayaz, M., Abdusalomov, A. B., & Cho, Y.-I. (2022). Effect of Feature Selection on the Accuracy of Music Popularity Classification Using Machine Learning Algorithms. <i>Electronics</i> , 11(21), 3518. https://doi.org/10.3390/electronics11213518	Inputs: 13 Spotify audio features (Danceability, Energy, Loudness, Valence etc.) Output: Popularity Class (High vs Low)	Compared SVM, Random Forest, Decision Tree	Used their findings to prioritize feature selection

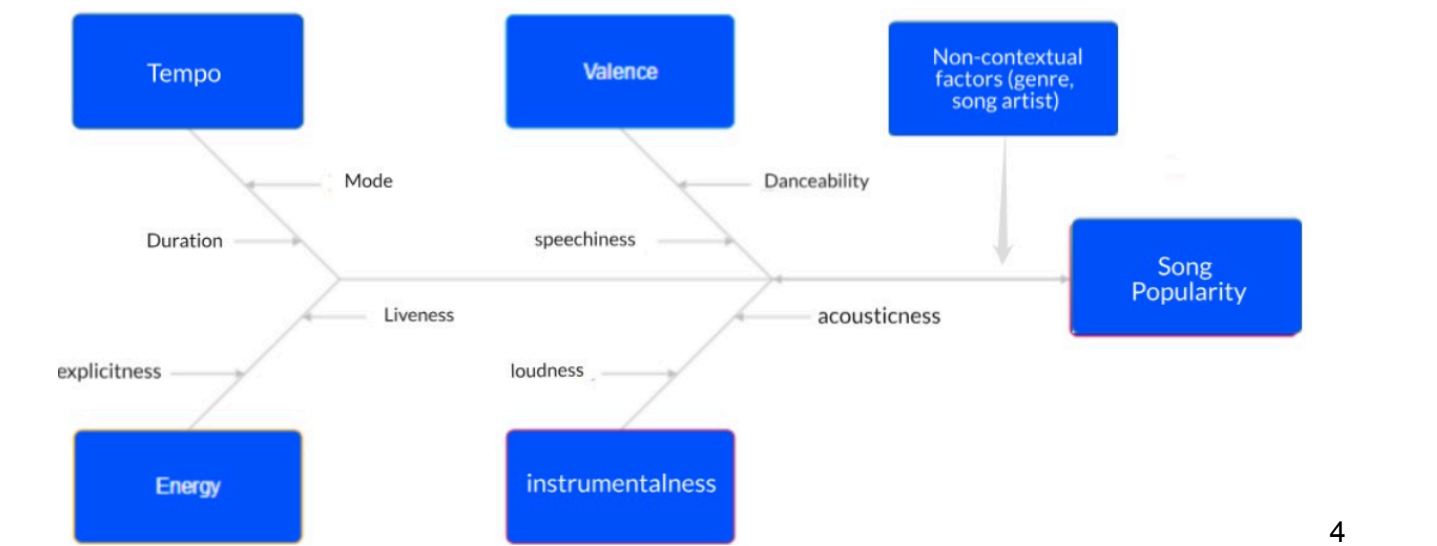
Li, K. (2024). Predicting song popularity in the digital age through Spotifys data. <i>Theoretical and Natural Science</i> , 39(1), 68–75. https://doi.org/10.54254/2753-8818/39/20240600	Inputs: Spotify audio features: (energy, danceability, speechiness, duration, mode); Output: Popularity (continuous)	Linear regression	We used their analysis to guide our normalization and scaling choices for continuous features prior to modeling.
Nijkamp, R. (2018). <i>Prediction of product success: explaining song popularity by audio features from Spotify data</i> . University of Twente. https://essay.utwente.nl/filesare/file/75422/NIJKAMP_BA_IBA.pdf	Inputs: 12 Spotify audio features (Liveness, Acousticness, Valence etc.) Output: # of Spotify Streams	Correlation Analysis and Multiple Linear Regression	We referenced their correlation analysis to verify linear relationships before modeling.
Wukkadada, B. (2025, January 25). <i>Decoding spotify hits: Statistical and predictive analysis of track features driving song popularity</i> . Academy of Marketing Studies Journal. https://www.abacademies.org/articles/decoding-spotify-hits-statistical-and-predictive-analysis-of-track-features-driving-song-popularity-17433.html	Inputs: 10 Spotify features (Valence, Tempo, Speechiness etc.) Output: Popularity Category (Hit vs Non-Hit)	Logistic Regression	Guided our approach to use supervised regression methods to predict continuous popularity scores based on Spotify audio features.

Figure 1: Table Summary of Related Research

Analytic Problem

The analytic problem we identified was which audio features contributed the most to a song's popularity. Our fishbone model shows the primary inputs based on audio features obtained from Spotify and Hot100. (Figure 2) Our modeling approach will evaluate how these audio features and metadata interact and determine which of them meaningfully impact popularity. Then, we will find out if the analysis tests which of these factors matter most for predicting song popularity.

Figure 2: Fishbone diagram



After our research and group discussion, we determined that a supervised learning approach would work best for this investigation, since our target variable of popularity is continuous and numerical. Our group thought that a regression model would be the most effective in telling us which features successfully determine the popularity of a song.

Data Description with Figures/Tables

Data Understanding

We were provided with four unique datasets that will help us answer our business question. The three datasets (large_set.csv, songs_csv and songs_csv) originated from the Spotify API, with its features generated from internal algorithms. “The popularity is calculated by algorithm and is based on the total number of plays the track has along with how recent those plays are.” ([Spotify Web API](#)) We found that the spotify_songs.csv file came from [tidytuesday 2020](#) and the others were likely online community data from sites like Kaggle. We then combined those datasets with a fourth dataset called the Hot100 which originates from historical data of every [Billboard Hot 100 chart](#) since 1958-2025. The dataset comes from online GitHub repo [HipsterVizNinja](#). The billboard dataset offered a more robust measure of sustained popularity compared to Spotify's momentary snapshot.

The datasets were all in csv (Comma-Separated Values) files which were easy to read and work with. In the Spotify-based datasets, each row represented a song and its audio features, metadata, and popularity. In the Billboard Hot 100 dataset, each row represented a chart position of a weekly Hot 100 chart with the associated song info and performance metrics. The initial characteristics of the combined raw data included over 500,000 Spotify records and 350,000 chart rankings. The quality of the data was good with minimal missing values showing strong completeness. The Hot 100 dataset did initially show a lot of missing values but that was due to a column named “previous_week”. This column was always empty when a song first hit the charts since there were no previous rankings. Analyzing the data without this column showed that there was no missing data for Hot 100. (Figure 3)

	Dataset	Row Count	Column Count	Missing Values	Missing Values (No previous_week)
0	spotify.csv	32833	23	15	NaN
1	songs.csv	81656	19	0	NaN
2	large_set.csv	396610	16	7	NaN
3	hot_100.csv	350583	13	35715	0.0

Figure 3: Table of initial dataset characteristics

Assumptions

We as a group did not collect the data ourselves, so we must make assumptions. For accuracy, coming from online sources, the Spotify datasets were a trusted source, making them relevant and reliable for this investigation. Similarly for Billboard Hot 100, the original source is trusted.

Doing a manual check showed no significant data accuracy issues. For completeness, we assume it is complete for our purposes. Technically it does not encompass all songs of all time and regions, but our data represents the USA and recently popular songs. We do not need to analyze songs from other regions and songs that were popular a long time ago are less likely to be relevant now. There is possible bias if certain genres or artists are overrepresented but that is the nature of popularity, so it is a non-factor. For ethics, we assume the data is ethical as it does not include any personally identifiable information of users, artists, or businesses. The datasets only pertain to song-based data like music features and popularity.

Looking at the data distribution of the datasets, most of the features were normally distributed. Some of them were shifted left or right. The only distribution that stood out was the release date vs popularity. In figure 4, the top row of plots shows that a lot of songs get released in January. We concluded that the discrepancy was due to song re-releases, business arrangements, or internal Spotify practices. Internal Spotify practices would include data errors, misinputs, or a default release date system. Looking at popularity across months did not have a strong visual variation in initial distribution. We did attempt to use Hot 100's release date as a replacement, but overall release date did not have a big impact on modeling later on so we ended up dropping it all together.

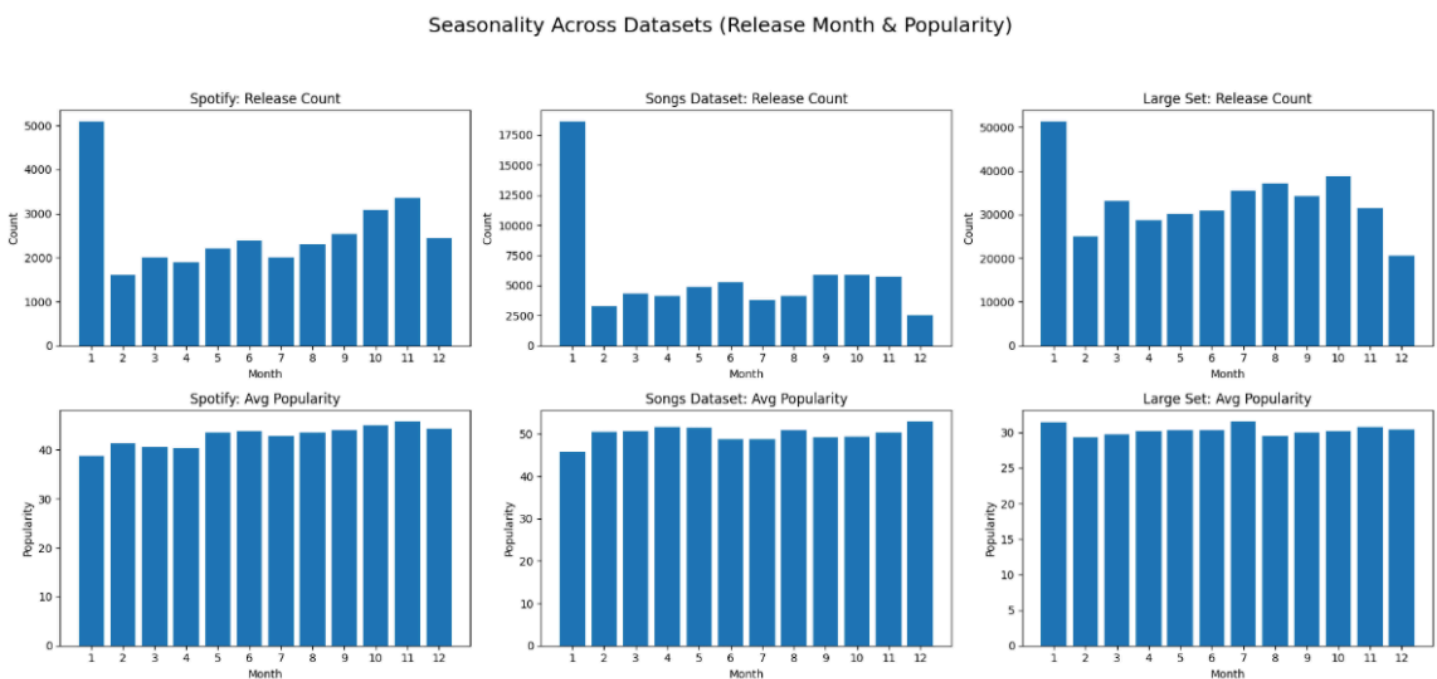


Figure 4: Seasonality plots of initial Spotify data.

The distribution of song popularity is heavily right-skewed with few songs making it past 20 weeks on chart, indicating that chart longevity is rare. The boxplot also shows this with a median duration of just 9 weeks and IQR of 12. (Figure 5) There are 552 outliers out of the 32287 total songs in the Hot 100 dataset.

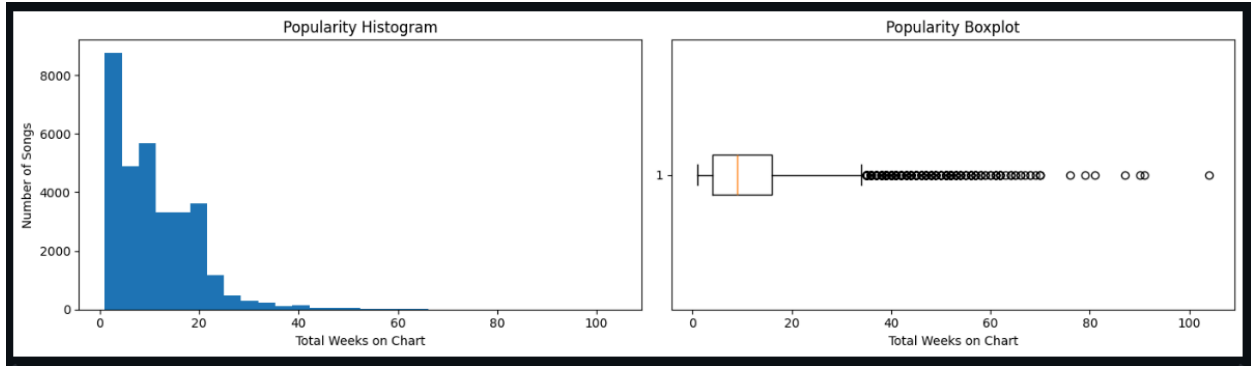


Figure 5: Distributions of Hot 100 Popularity.

Input Selection & Brainstorming Summary

Analysis and Research

To determine which inputs to use, we reviewed all the columns that appeared in the merged Spotify datasets and the Hot 100 dataset. We focused only on features that: appear consistently across datasets, are numeric and ready for modeling, and directly relate to our business question about audio features and popularity.

We selected the following inputs Danceability, Energy, Speechiness, Liveness, Acousticness, Instrumentalness, Loudness, Valence, Duration, Tempo, Mode, Explicit. These inputs were chosen because they are the standardized Spotify audio features. We did not include identifiers, text fields, or duplicate naming columns because they do not help predict popularity.

Black Box Diagram and Discussion

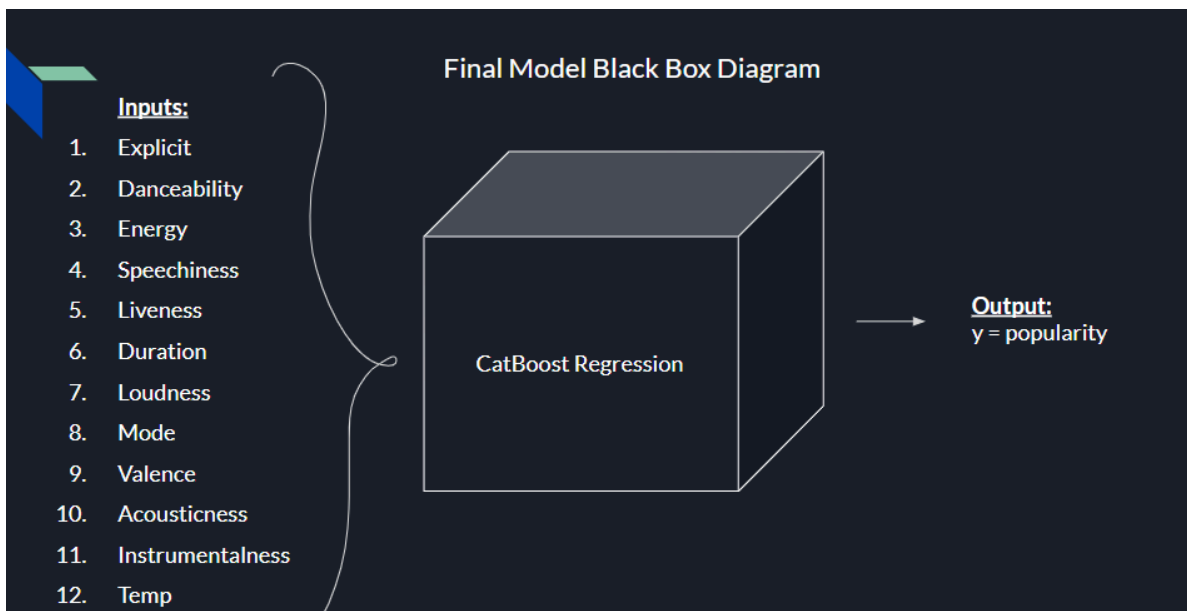


Figure 6: Final Model Black Box Diagram

The final model we went with and developed for this project was the CatBoost Regression model. It was selected for its ability to handle nonlinear relationships and mixed feature types without extensive preprocessing. The model uses twelve quantitative input variables extracted from the Spotify dataset: *explicit*, *danceability*, *energy*, *speechiness*, *liveness*, *duration*, *loudness*, *mode*, *valence*, *acousticness*, *instrumentalness*, and *tempo*. Each of these features represents a measurable audio feature where the algorithm analyzes combined influence to generate the output of a predicted popularity score (y). Our black box diagram visually summarizes the full modeling process by labeling the inputs, the model type, and the output variable.

Methodology Diagram

Data Preparation

We started by merging the three Spotify-based datasets. This reduced over 500,000 raw records to 3838 rows after removing duplicates and irrelevant columns such as IDs, repeated name fields, playlist information, and other non-predictive metadata. Things like song name, song id, track id, playlist id, album id, etc. Continuous features were standardized because some datasets used a 1-100 scale while others used 0-1. This prevents large-scale variables like tempo from dominating model coefficients.

Integrating Hot 100 data

We merged the Billboard Hot 100 dataset to add “time_on_chart”, a more stable measure of popularity. This corrects for Spotify’s algorithmic popularity score, which can spike temporarily and does not capture long-term success.

	Variable	Data Type	Num Missing	Value Ranges	Example Values	Description
0	explicit	int64	0	0 – 1	[0, 1]	Explicit content flag (0 = No, 1 = Yes)
1	popularity	float64	0	1.0 – 87.0	[47.0, 23.0, 18.0]	Popularity score (0–100)
2	danceability	float64	0	0.178 – 0.979	[0.385, 0.619, 0.542]	Danceability score (0–1)
3	energy	float64	0	0.077 – 0.997	[0.586, 0.491, 0.466]	Energy score (0–1)
4	speechiness	float64	0	0.022 – 0.578	[0.033, 0.032, 0.028]	Speechiness score (0–1)
5	liveness	float64	0	0.02 – 0.943	[0.186, 0.107, 0.133]	Liveness score (0–1)
6	duration	float64	0	120133.0 – 431800.0	[256853.5, 305827.0, 295871.5]	Track duration in milliseconds (ms)
7	loudness	float64	0	-18.999 – -1.653	[-7.55, -9.688, -10.328]	Loudness level in decibels (dB)
8	mode	float64	0	0.0 – 1.0	[1.0, 0.5, 0.0]	Mode (1 = Major, 0 = Minor)
9	valence	float64	0	0.035 – 0.98	[0.146, 0.553, 0.118]	Valence (positivity/mood, 0–1)
10	acousticness	float64	0	0.0 – 0.965	[0.018, 0.455, 0.0]	Acousticness score (0–1)
11	instrumentalness	float64	0	0.0 – 0.837	[0.0, 0.0, 0.295]	Instrumentalness score (0–1)
12	tempo	float64	0	57.978 – 200.204	[139.942, 131.925, 110.03]	Tempo of the track in beats per minute (BPM)

Figure 7: Final Data Dictionary Table

Feature Selection

We kept all Spotify audio features that were numeric, consistent, and meaningful for modeling: Danceability, Energy, Speechiness, Liveness, Acousticness, Instrumentalness, Loudness, Valence, Duration, Tempo, Mode, Explicit. Combined with the Hot 100 popularity, we ended up with a final dataset of 1805 rows and 13 columns. (Figure 7)

Train-Test Split

We split the dataset into 70% training and 30% testing to evaluate performance on unseen data.

Baseline modeling

Three baseline regression models were trained using default parameters with 5-fold cross validation: Random Forest, XGBoost, CatBoost. CatBoost showed the strongest baseline performance with the least overfitting.

Hyperparameter tuning

We used RandomizedSearchCV with a 5-fold CV to tune each model. CatBoost again performed the best with improved generalization and lower error.

Final Modeling

The final model (CatBoost) was retrained using the best parameters. It explained around 73-76% of the variance in popularity ($\text{Train } R^2 \rightarrow \text{Test } R^2$) and had typical errors within 7-9 popularity points.

Model output

The model outputs a predicted popularity score based on a song's audio features

Our full modeling pipeline is summarized in Figure 8, which outlines each step from data preparation to final model output.

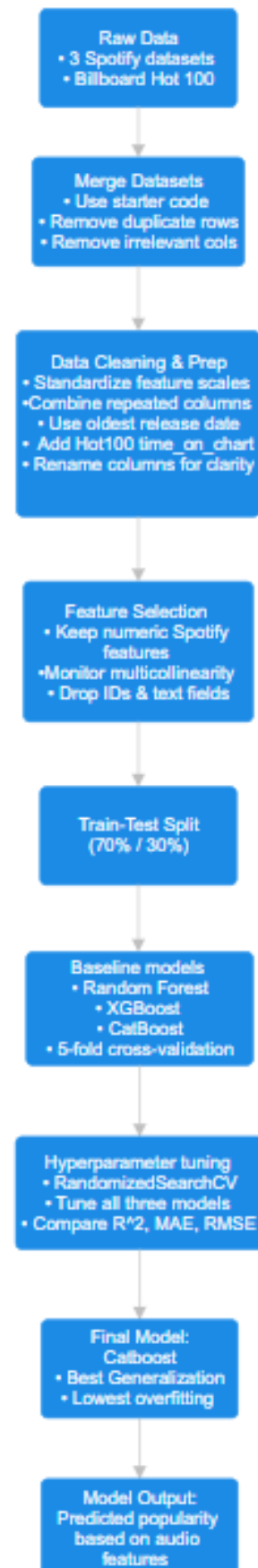


Figure 8: Methodology Pipeline

Implementation & Analysis

Evaluation Setup and Metrics

To evaluate our models fairly, we first split the merged datasets into 70% for training and 30% for testing, resulting in 1,263 training songs and 542 test songs. All tuning decisions were made using the training data only. Within the training set, we used 5-fold K-Fold cross-validation, where the model is trained on four folds and validated on the remaining fold, and this process is repeated five times. This approach reduces variance in our performance estimates and makes our conclusions less sensitive to how the data happened to be split.

The target variable popularity is continuous, so we framed the task as a regression problem. We monitored four key metrics: Train R^2 and Test R^2 to measure how much variance is explained; MAE to show the average prediction error in popularity points; and RMSE to highlight larger errors. We also computed an overfit gap (Train R^2 - Test R^2). This single number helped us quickly see whether a model was overfitting (large positive gap) or achieving similar performance on unseen songs (small gap).

Baseline Model Implementation

As a first modeling step, we trained three baseline regressors using default hyperparameters: Random Forest, XGBoost, and CatBoost. These models are well-suited for problems with weak linear relationships and potential non-linear effects, which aligns with characteristics observed in our dataset. Using default parameters provided a neutral starting point to assess each model's natural behavior on the data.

All baseline models were evaluated using 5-fold cross-validation on the training set. After cross-validation, each model was retrained on the full training split and tested on the unseen test set. Performance was measured using R^2 , MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and the overfit gap. The results showed that although Random Forest achieved the highest training R^2 , CatBoost delivered the most consistent performance between training and testing, indicating better generalization at the baseline level (Table 9.1).

Random Forest Regressor (Default Parameters and KFold CV)	→	<pre> ===== Random Forest Regressor (Default Parameters) ===== Train R² : 0.7998 Test R² : 0.7274 Test RMSE : 10.428 Test MAE : 8.315 Overfit Gap : 0.0724 ===== </pre>
XGBoost Regressor (Default Parameters and KFold CV)	→	<pre> ===== XGBoost Regressor (Default Parameters) ===== Train R² : 0.7805 Test R² : 0.6873 Test RMSE : 10.612 Test MAE : 8.489 Overfit Gap : 0.0932 ===== </pre>
CatBoost Regressor (Default Parameters and KFold CV)	→	<pre> ===== CatBoost Regressor (Default Parameters) ===== Train R² : 0.7513 Test R² : 0.7195 Test RMSE : 10.612 Test MAE : 8.201 Overfit Gap : 0.0318 ===== </pre>

Table 9.1: Baseline Model Performance Comparison (Train/Test R^2 , MAE, RMSE, Overfitting Gap)

Hyperparameter Tuning Strategy

To enhance predictive performance and better control overfitting, we carried out hyperparameter tuning with `RandomizedSearchCV` using 5-fold cross-validation. This approach allowed us to efficiently test many different parameter combinations while avoiding the heavy computational load of exhaustive search. All tuning was restricted to the training set to ensure that our final test results remained unbiased.

For each model, we defined a reasonable tuning range based on standard model behavior. Random Forest tuning focused on the number of trees and tree depth. XGBoost tuning targeted the learning rate, number of estimators, depth, and subsample ratio. CatBoost tuning focused on the number of boosting iterations, learning rate, depth, and L2 regularization term. The best hyperparameters for each model were selected according to the highest mean cross-validation R^2 , and these were used in the final model refitting phase (Table 9.2).

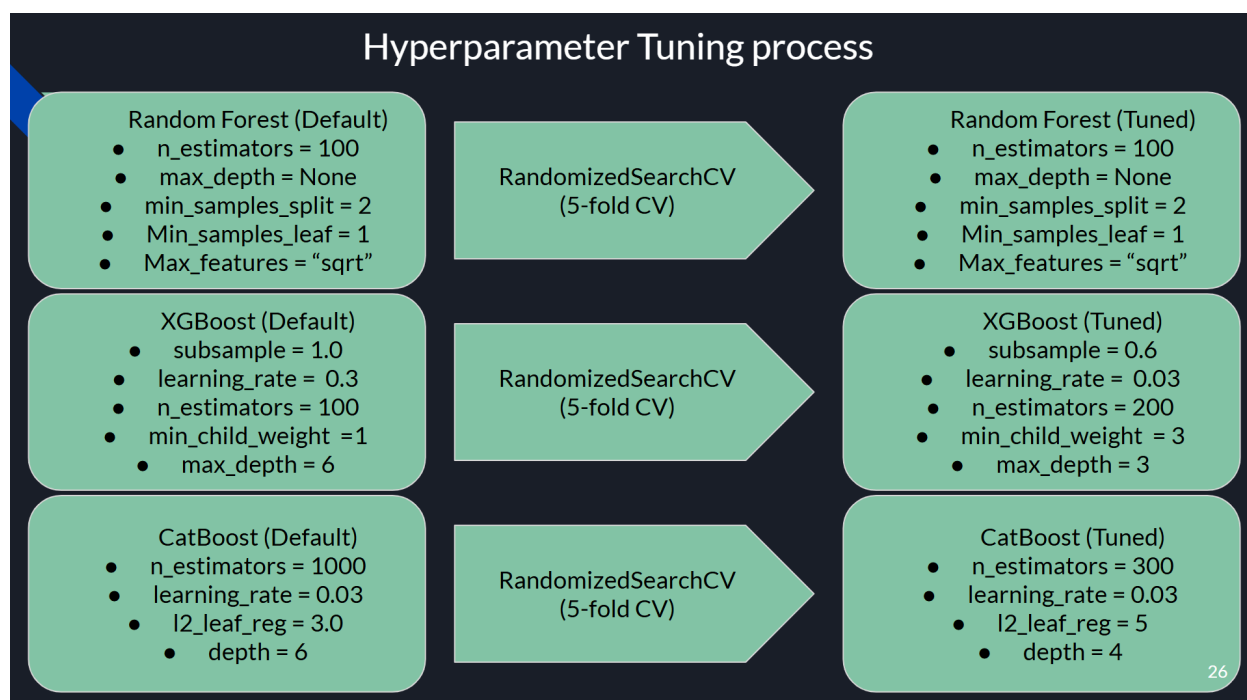


Table 9.2: Hyperparameter Tuning Process

Tuned Model Performance and Overfitting Analysis

With the optimized hyperparameters from RandomizedSearchCV, each model was retrained on the full training set and evaluated on the test set. Comparing tuned models to their baseline counterparts showed that tuning generally improved predictive performance, particularly in reducing typical error magnitudes measured by MAE and RMSE.

For the Random Forest (tuned) model, test error decreased slightly, but Train and Test R^2 remained relatively far apart. This indicates limited improvement in controlling overfitting (Table 9.3).

The tuned XGBoost model showed better generalization than its baseline version. Its Test R^2 increased and overfit gap decreased, reflecting the benefits of stronger regularization and a lower learning rate (Table 9.3).

The tuned CatBoost model achieved the strongest performance across all metrics. It produced the highest Test R^2 , the lowest MAE and RMSE, and the smallest Train-Test gap. These results demonstrate that CatBoost provided the most reliable balance between model complexity and generalization capability (Table 9.3).

Random Forest Regressor (Tuned Parameters and KFold CV)	→	<pre> ===== Random Forest Regressor (Tuned Parameters) ===== Train R² : 0.7857 Test R² : 0.7174 Test RMSE : 9.148 Test MAE : 7.321 Overfit Gap : 0.0683 ===== </pre>
XGBoost Regressor (Tuned Parameters and KFold CV)	→	<pre> ===== XGBoost Regressor (Tuned Parameters) ===== Train R² : 0.7913 Test R² : 0.7152 Test RMSE : 9.285 Test MAE : 7.456 Overfit Gap : 0.0761 ===== </pre>
CatBoost Regressor (Tuned Parameters and KFold CV)	→	<pre> ===== CatBoost Regressor (Tuned Parameters) ===== Train R² : 0.7593 Test R² : 0.7381 Test RMSE : 8.971 Test MAE : 7.182 Overfit Gap : 0.0212 ===== </pre>

Table 9.3: Tuned Model Performance Comparison

What Worked, What Did Not, and Why

Our results show that using non-linear models was essential for predicting song popularity from audio features. Linear relationships alone were not sufficient, and tree-based models were able to capture interactions between features. This allowed all three models to achieve reasonably strong predictive performance.

The CatBoost model emerged as the most balanced and robust. It produced the highest test R^2 and the lowest error values, while also maintaining the smallest overfitting gap. This suggests that CatBoost's regularization mechanisms and boosting strategy were well-suited to the structure of our data.

In contrast, XGBoost was initially too flexible. Its baseline version showed clear signs of overfitting, with high training accuracy and weaker test performance. Tuning improved this behavior, but it still required careful parameter control to avoid memorization.

A major limitation that affected all models was the relatively small dataset size. With only 1,805 observations, there is a natural cap on how much variance can be explained. More data across additional genres, time periods, or artists could significantly improve future model performance.

Summary

Overall, our implementation strategy combined a consistent evaluation setup, three strong ensemble baselines, and targeted hyperparameter tuning. While all models benefited from this pipeline, CatBoost delivered the best balance of high Test R^2 , low MAE/RMSE, and minimal overfitting. Random Forest and XGBoost were competitive but either showed less improvement with tuning or required stronger regularization to avoid memorizing noise. These findings set the stage for the next section, which focuses on interpreting the final model's performance and tells us about audio features and popularity.

Results & Interpretation

Evaluation

After testing the final model using Catboost, we found that loudness and acousticness have a reasonable audio correlation. By dropping Release Month and Key, had helped our model reduce overfitting. The R^2 explained between 73% trained variation and 76% test variation of the variation in popularity, improving the model by ~ 0.02 , and reducing the overfit gap from ~ 0.0318 to ~ 0.0212 .

Conclusion & Future Work

Deployment

We had come to the conclusion that our model could have had better performance. Regarding the question of alignment or not, our results were partially aligned. The results did answer our business question by focusing on the insights of popularity; however, our question was more about the audio features emphasizing a song's popularity. Our model would be useful if deployed, but popularity is affected by other external factors not included in our model. Other external factors would be marketing (ads, social media campaigns), existing artist fame and influence, or Spotify recommendation algorithm bias. To better predict popularity, we have to incorporate non-audio data into our model. Additionally, popularity will shift and grow as time goes on. Music trends come and go, and our model needs to be built to fetch and process new data.

AI Disclosure: AI tools were used to assist with coding.

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